

HABIT LOOP MIRROR

"Screen Time tells you how much. We tell you why."

Understand the patterns behind your digital habits.

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1. WHY

What is the real-world problem?

Phones show screen time as a number: hours per day, minutes per app. That says how much, but not why. Someone can see "three hours on one app" and still not know whether it was one planned sitting or many short, unplanned opens through the day, or what led to each one. A number like that is easy to feel bad about and hard to act on.

Why did I choose this problem?

Existing screen-time tools stop at measurement. What they leave out is the pattern behind the numbers: when sessions start, what usually comes just before them (a notification, for example), and whether the person meant to open the app at all. I wanted to build something that answers "why" instead of repeating "how much".

Why did I choose SDG 3?

SDG 3, Good Health and Well-Being, is about ensuring healthy lives and promoting well-being for all at all ages, and Target 3.4 includes promoting mental health and well-being. Habit Loop Mirror connects to this through everyday digital well-being: helping people notice their habits and use their devices more intentionally. It is not a healthcare tool and does not diagnose or treat anything.

Why is solving this problem important?

Apps are part of studying, working and relaxing, so the goal is not to stop using them but to use them on purpose. That needs three things: noticing your own pattern, thinking about it, and having something better to do instead. Reports that only show large totals tend to feel like criticism. A plain explanation without blame gives people something they can work with.

2. WHAT

What is Habit Loop Mirror?

It is a web platform that takes a person's digital usage data, finds the habits that keep recurring, explains them in plain language, and suggests alternatives. It is an awareness and self-regulation tool, not a medical or diagnostic one.

How does it solve the problem?

It works as a five-stage loop: Detect → Explain → Reflect → Substitute → Improve.

Stage	What happens
1. Detect	Matches app notifications to the time each session started, and looks for the same pattern repeating across several days.
2. Explain	Shows the pattern as Trigger, Action, Latency (how long after the trigger the app was opened) and Duration (how long the session lasted). The wording is neutral.
3. Reflect	The user labels each session as Planned, Necessary, Relaxation or Unplanned.
4. Substitute	Suggests alternatives of 2, 10 and 20 minutes, based on what the user chose during onboarding.
5. Improve	Tracks the pattern over several weeks and works out time reclaimed from recorded data only.

What are its main features?

- Analysis of digital usage data and detection of recurring habit loops
- Explanations broken into trigger, action, latency and duration
- Session reflection (planned, necessary, relaxation, unplanned)
- Personalised 2, 10 and 20-minute alternatives
- Habit evolution tracking over weeks, with reclaimed time

Where and how is AI being used?

There are two layers. The analytics layer uses Python, Pandas, NumPy and Scikit-learn to find the patterns, and this is the evidence the rest of the system relies on. The AI layer connects to Kimi K3 through a Moonshot AI adapter and turns those findings into readable insights. It explains what the analytics found and does not add claims of its own. If the AI service is unavailable, a rule-based fallback produces insights from the same analytics. The platform is built with FastAPI, React and TypeScript, with PostgreSQL / SQLite and Supabase for storage. I am not reporting any accuracy figure.

3. WHOM

Who are the target users?

College students, young adults, and anyone who uses digital platforms often and wants to understand their own habits better. The first users I expect are students who often open apps without planning to, and people interested in productivity and self-awareness.

Who benefits, and what is the specific benefit?

- **Students and young adults:** see which repeated patterns they have, instead of one total.
- **Users who open apps unplanned:** can label sessions and see how often each type happens.
- **Users who want a change:** get alternatives sized to the time they have (2, 10 or 20 minutes).
- **Users who stay with it:** can watch their habits change over several weeks.

Habit Loop Mirror is meant to support these outcomes. No results have been measured yet, and it does not claim to change behavior on its own.

Conclusion

Screen-time numbers answer “how much” but leave users without an explanation. Habit Loop Mirror adds that explanation through a five-stage loop, backed by data analytics and written up with AI, for students, young adults and frequent digital users. In doing so it supports the well-being side of SDG 3, by helping people understand and shape their digital routines.